

Fertility and housing: evening calibration

What improved, what remains missing, and the complete target fit

The run completed, and fit improved: the best common-primary score fell from 33.821 to **26.368** (22.035%). The point comes from the block-weight search. It is a reproducible candidate, not an established optimum.

Scope: 24 single-thread Torch workers; 360 search attempts: 221 accepted, 77 housing-equilibrium gate rejections, 61 owned timeouts and one late completion excluded. No fatal error; six fresh final repeats passed. The scientific contract stayed fixed during the accepted search; no local model computation.

Search / reference	Score under the same primary weights
Starting point	33.821
Primary weights	26.405
Relative-gap identity weights	113.584
Equal-block weights (common best)	26.368

All 14 rows: ten scored moments, three validations, one normalization

Moment	Target	Model	Gap	Weight	Loss
Completed fertility (normalization)	2.1	2.1	2.18e-04	--	--
Childlessness, ages 40-44	0.198	0.2	0.001	3.55e+04	0.079
Exactly one child, among mothers	0.214	0.211	-0.002	2.70e+04	0.152
Mean age at first birth	25.976	25.984	0.007	139.828	0.007
First births at age 30+ [V]	0.249	0.227	-0.022	0	0
Aggregate wealth / annual earnings	6.927	6.23	-0.697	7.595	3.688
Annual bequest flow / wealth	0.007	0.007	-2.33e-04	5.17e+06	0.28
Older wealth/income: p90 / median [V]	3.516	3.114	-0.402	0	0
Mean occupied rooms	5.729	5.894	0.164	128.021	3.46
Ownership, ages 30-55	0.676	0.633	-0.043	2339.362	4.295
Housing response to first birth	1.465	1.673	0.208	137.565	5.947
Rooms: 3+ vs 1-2 children at home [V]	0.385	0.314	-0.071	0	0
Recent-parent ownership gap	0.128	0.122	-0.006	2.71e+04	0.867
Children ever born at 25, capped at 3	0.81	0.534	-0.276	100	7.595

Gap = model - target. All weights and loss contributions above are primary weights, including zero weights for the three untargeted validation rows [V]. Completed fertility fixes the benefit level and is not scored. Identity and block objective scalars are never directly compared.

The main economic misses persist: early fertility is 0.534 versus 0.810; ownership is 63.341% versus 67.626%; wealth/earnings is 6.230 versus 6.927. Housing rises too much at first birth (1.673 versus 1.465 rooms). Childlessness, one-child frequency and mean first-birth age are close to their targets.

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Parameter estimates, numerical confidence, and the next decisions

Eleven fitted quantities: ten searched parameters, including the tenure-choice scale, plus the child-benefit level normalized to completed fertility 2.1. Ten scored moments plus that normalization give eleven restrictions by count; identification is not established.

Parameter	Estimate	Lower	Upper	Bound / restriction
Housing supply level	6.206	0.2	80	interior
Annual discount factor	0.963	0.94	0.99	interior
Owner housing-service premium	1.089	0.1	5	interior
First-birth fixed cost	0.647	0	8	interior
First-birth taste scale	0.187	0.02	50	near lower
Later-birth taste scale	0.347	0.02	50	near lower
Bequest strength	0.114	0	8	interior
First-child housing loading	0.142	0	0.25	interior
Child-benefit curvature	0.079	0	0.8	interior
Tenure-choice taste scale	0.008	0.001	0.1	interior
Child-benefit level (normalized)	0.141	--	--	positive; normalized

Near-bound flags use 1% of the full search interval; neither fertility taste scale hits its lower bound. The remaining fixed/derived parameters are preserved in the full 31-row CSV. Annual real interest stays 2%; the existing-owner borrowing rule and separate death-solvency condition are the author-adopted DUE specification.

What the numerical evidence supports

Each of the three selected candidates has two fresh successful repeats with identical full 14-row target and 31-row parameter tables. All 17 figures per lane match repeat sources and passed final visual review. The common best has relative housing-market residual 4.84e-07 and completed-fertility error 2.18e-04. The cluster job ended normally; peak memory was about 102.1 GiB of 128 GiB.

Limits that matter for interpretation

The first-birth housing contrast is a four-year model comparison rather than the exact empirical event-study estimator. Bequest data measure child-directed transfers while the model counts positive estates. The three zero-weight rows remain visible validation comparisons. The housing/fertility tradeoff is therefore a mechanism question and a measurement question.

The standard policy plots are buyer-conditional: they do not certify every occupied stayer policy. High-wealth policy downturns, late ownership and retirement decumulation remain diagnostic caveats. Timeouts and gate rejections limit the explored region; a finite search does not establish an optimum or an identification rank.

Proposed next decisions - no automatic continuation

- Keep the same targets for a longer, better-initialized search; diagnose the 900-second timeouts without relaxing equilibrium gates.
- Measure local sensitivity around this candidate, including tenure scale, housing loading and fertility preferences, to assess parameter substitution.
- Review the early-fertility / first-birth-housing tension and the provisional data counterparts; retain the three validation rows throughout.

Supporting evidence: final_report/summary.json, common_best_target_fit.csv and per-lane full parameter/fit CSVs; the stable 17 figures remain separate. No transition result or policy conclusion is claimed.